

Original article

Rapid oscillatory activity in delta brushes of premature and term neonatal EEG

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Abstract

We compared frequency and power of neonatal EEG delta brush rapid oscillatory activity (ROA) using multiple band frequency analysis (MBFA) in three groups; pre-term (PT, post-conceptual age 33–35.6 weeks, $n = 5$); full-term (FT, 39.4–40.6 weeks, $n = 5$) and pre-term or full-term with phenobarbital exposure (PB, $n = 5$). Mean number of delta brushes analyzed was 29.4 (range 26–47) in PT, 20.8 (14–33) in FT and 20 (7–37) in PB. Mean frequency $\pm$ standard deviation (s.d.) was 16.9 ± 2.1 Hz (range 15–20 Hz) in PT, 17.3 ± 1.9 Hz (15–20 Hz) in FT and 16.1 ± 1.6 Hz (14–19 Hz) in PB. Mean power $\pm$ s.d. was $22.9 \pm 6.2 \mu\text{V}^2$ (range 16–39 μV^2) in PT, $11.9 \pm 4.1 \mu\text{V}^2$ (7–19 μV^2) in FT and $17.1 \pm 6.2 \mu\text{V}^2$ (9–26 μV^2) in PB. Power was significantly higher in PT than FT ($p < 0.005$). Power after merging PB into respective PT (PT', $n = 8$) and FT (FT', $n = 7$) groups, remained significantly higher in PT' (mean $\pm$ s.d. $21.8 \pm 7.4 \mu\text{V}^2$) than FT' ($11.4 \pm 3.6 \mu\text{V}^2$) ($p < 0.05$). We characterise ROA in delta brushes in maturing neonates using MBFA, which may provide additional information for assessing future seizure recurrence and epilepsy risk.

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1. Introduction

Delta brushes are age-related EEG waveforms found in neonatal EEG recordings. They comprise epochs of 8–22 Hz rhythmic fast activity of amplitude 20–200 μV superimposed on 0.3–1.5 Hz slow activity of amplitude 50–250 μV . They have a duration of 0.3–1.2 s and have a predominant occipital, central and temporal distribution [1]. Delta brushes are rarely seen at 24–25 week post-conceptual age but increase with maturity. They become most prominent and frequent by 30–32 weeks post-conceptual age then become less evident and

are rare by 40 weeks post-conceptual age [2]. Up to 33–34 weeks post-conceptual age they are more commonly seen in the awake and active sleep recording than quiet sleep. From 35 weeks they become more evident in quiet sleep compared to active sleep.

Multiple band frequency analysis (MBFA), like the fast Fourier Transform and wavelet transform, is a method of frequency analysis. It was developed to analyze spectral density characteristics in visual evoked potentials [3]. MBFA separates the raw signal into multiple sub-band signal components by utilizing multiple band-pass filters.

Rapid oscillatory activity (ROA) can be a normal physiological phenomenon and appears from 6 months post-conceptual age in neonates. ROA is seen during motor activity and is reported to be involved in somatopic organization [4]. ROA is present in the delta

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brushes of the normal neonatal EEG. Multiple mechanisms may modulate oscillatory activity in the brain including gap junctions, calcium channels and gamma amino butyric acid-A (GABA_A) receptors [5]. We performed objective quantitative analysis to determine the normal baseline characteristics of ROA in neonatal delta brushes and the effect of phenobarbital exposure. We hypothesize that we can identify differences in ROA in the maturing neonate using MBFA and establish baseline values to help assess risks for future seizure recurrence and developing epilepsy.

2. Method

We retrospectively selected neonates from the neurophysiology and EEG database at the Hospital for Sick Children, Toronto, Canada from January 2000 to January 2007. Inclusion criteria were neonates (1) of post-gestational age of 24 weeks or more (2) with EEG performed at conceptional age of 33–42 weeks (3) with EEG reported as normal and (4) with delta brushes seen on EEG. Neonates were excluded if they had (1) electro-clinical or electrographic seizures or (2) significant central nervous system pathology such as intracranial hemorrhage, hypoxic–ischemic encephalopathy, strokes, meningitis or encephalitis. Neonates with a history of apnea and/or non-neurological systemic illness were eligible for inclusion in the study. Neonates who had received phenobarbital empirically for suspected seizures and had a normal EEG were also eligible.

Patients were divided into 3 groups. The pre-term group (PT) were premature neonates of 24–34 weeks gestational age at birth and 33–37 weeks post-conceptional age at the time of EEG who had not received phe-

nobarbital therapy. The full-term group (FT) were term neonates of 38 weeks or more post-conceptional age who had not received phenobarbital therapy. The phenobarbital-treated group (PB) were pre-term and term neonates who had received phenobarbital therapy for suspected seizures but had no proven seizures at the time of EEG and had a normal EEG.

EEG records of eligible patients were extracted from archived recordings. EEGs were performed on digital EEG machines (XLTech, Oakville, Canada) utilizing 9 electrodes in the neonatal montage with a recording duration of 40–60 min with awake, quiet and active sleep states and a sampling rate of 500 Hz. We identified segments containing delta brushes from the posterior lead derivations (C3-O1, C4-O2, T3-O1, T4-O2) using low frequency filter 0.5 Hz, high frequency filter 70 Hz and notch filter 60 Hz. We exported this data as unfiltered EDF+ format files. The EEG segments were subjected to MBFA using Short Spectrum Eye software (Gram, Saitama, Japan) and EEG wave forms were deconstructed into serial frequency bands. The peak power at frequencies between 5 and 300 Hz was analyzed with 2 Hz resolution, which was equivalent to 147 serial band-pass filters. The amplitude of each frequency band was estimated using a 50 ms time window and 25 ms overlap. The amplitude of each frequency was measured using MBFA to calculate the power μV^2 . The calculated power of each frequency was expressed as a chart of power spectra during a time window (Fig. 1).

MBFA power values and frequencies of the delta brushes in each group were exported to SPSS (version 11.5, SPSS Inc., Chicago) and analyzed using one-way ANOVA and Tukey's test. Data on phenobarbital-exposed patients (PB) was divided into pre-term and

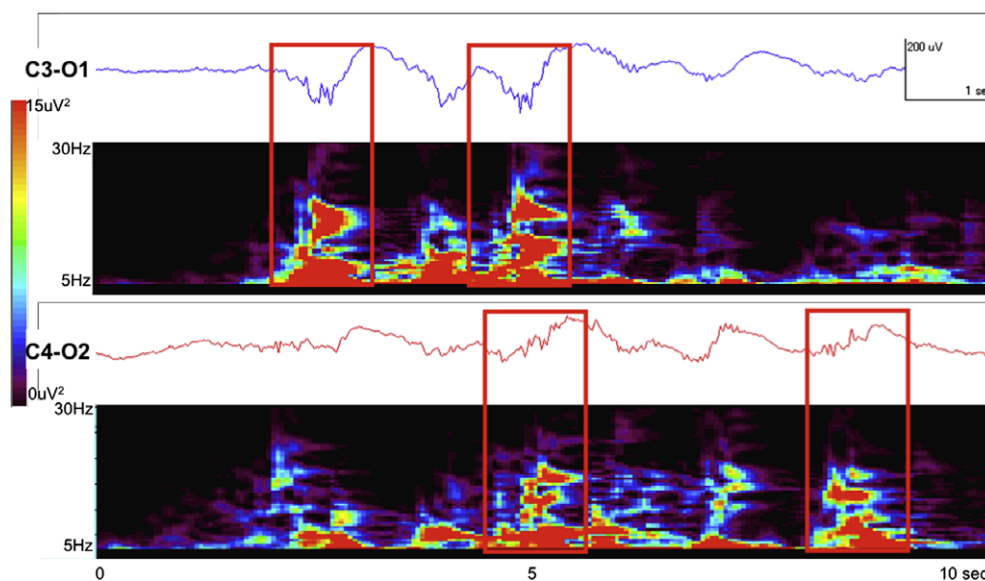


Fig. 1. Power spectrogram (5–30 Hz, 10 s) of EEG excerpts with delta brush (C3-O1, C4-O2) using multiple band frequency analysis (MBFA). Delta brushes (boxed) show high power across low and high frequency bands.

term groups and compared using independent *t*-tests. Expedited review and approval was obtained from the hospital research ethics board.

3. Results

We identified 15 neonates with delta brushes, 5 in each group. Post-conceptual age at the point of EEG recording was PT: 33–35.7 weeks (mean 34.8 weeks), FT: 39.4–40.7 weeks (mean 40.3 weeks) and PB: 31.6–40.6 weeks (mean 36.3 weeks) and is listed in Table 1. The number of delta brushes analyzed in each patient ranged from 7 to 37. Eleven neonates each had 10 epochs with delta brush activity. Three patients in the FT group had 6–8 epochs and one patient in the PB group had 3 epochs. Forty-one to 50 epochs were analyzed in each group. The mean number of delta brushes in the PT group was 33.4 (range 26–47), which was significantly higher than the FT [mean 20.8 (range 14–33)] and PB [mean 20 (range 7–37)] groups ($p < 0.05$). Table 1 describes the characteristics of patients and delta brushes in each group. The PB group had received phenobarbital in doses ranging from 2 to 5 mg/kg/day and had been on phenobarbital from 2 to 15 days prior to EEG recording.

Table 2 describes the mean frequency and power in each group. Mean frequency $\pm$ standard deviation (s.d.) was 16.9 ± 2.1 Hz (range 15–20 Hz) in the PT group, 17.3 ± 1.9 Hz (15–20 Hz) in the FT group and 16.1 ± 1.6 Hz (14–19 Hz) in the PB group. Mean power $\pm$ s.d. was $22.9 \pm 6.2 \mu\text{V}^2$ (range 16–39 μV^2) in the PT group, $11.9 \pm 4.1 \mu\text{V}^2$ (7–19 μV^2) in the FT group and $17.1 \pm 6.2 \mu\text{V}^2$ (9–26 μV^2) in the PB group. Power was significantly higher in the PT group compared to the FT group on post-hoc analysis with Tukey's test ($p < 0.005$).

Higher amplitudes were seen in the PT group compared to the FT group. No significant difference was seen in the phenobarbital-exposed group compared to both PT and FT groups.

The frequency and power values of the PB group were similar to those in the respective untreated PT and FT groups (Table 3). Pre-term patients in the PB group were merged with the PT group and renamed PT'. Full-term patients in the PB group were merged with the FT group and renamed FT'. Post-conceptual age range at the time of EEG was 31–34 weeks (mean 34.3 weeks) in PT' and 39–41 weeks (mean 40.1 weeks) in FT'. Power remained significantly higher in the pre-term group; PT' (mean $\pm$ s.d. $21.8 \pm 7.4 \mu\text{V}^2$) than FT' ($11.4 \pm 3.6 \mu\text{V}^2$) ($p < 0.05$).

Table 1
Summary of demographics, EEG and delta brush characteristics of subjects.

	Gestational age at birth	Post-conceptual age at EEG	Delta brush number	Mean delta brush number	
Pre-term	25	35.6	28	33.4	} $P < 0.05$
	29	34.8	39		
	30.6	33	27		
	34.3	35.7	47		
	26.7	34.7	26		
Full-term	40	40	33	20.8	
	39	39.4	13		
	40	40.4	14		
	40	40.6	14		
	40	40.3	30		
Phenobarbital	35	36	22	20	
	40	40.6	7		
	31	31.6	37		
	40	40.2	11		
	31	33.1	23		

Table 2
Frequency and power of rapid oscillatory activity of delta brushes in pre-term (PT) full-term (FT), and phenobarbital (PB) groups.

Group	Number of subjects	Frequency (Hz)		Power (μV^2)	
		Mean $\pm$ s.d.	Range	Mean $\pm$ s.d.	Range
PT	5	16.9 ± 2.1	15–20	22.9 ± 6.2	} $P < 0.005$
FT	5	17.3 ± 1.9	15–20	11.9 ± 4.1	
PB	5	16.1 ± 1.6	14–19	17.1 ± 6.2	

Table 3

Frequency and power of rapid oscillatory activity of delta brushes in pre-term (PT') and full-term (FT') groups with and without phenobarbital.

Merged group	Number of subjects	Mean age at EEG (weeks)	Frequency (Hz) mean $\pm$ s.d.	Power (μV^2) mean $\pm$ s.d.	} $P < 0.05$
PT'	8	34.3	16.6 $\pm$ 1.8	21.8 $\pm$ 7.4	
FT'	7	40.1	17.2 $\pm$ 1.9	11.4 $\pm$ 3.6	

In the PT' group, ROA of delta brushes had mean frequency 16.6 Hz (s.d. $\pm$ 1.8 Hz, range 15–20 Hz) and mean power 21.8 μV^2 (s.d. $\pm$ 7.4 μV^2 ; range 16–40 μV^2). In the FT' group, ROA had frequency 17.2 Hz (s.d. $\pm$ 1.9 Hz; range 15–20 Hz) and mean power 11.4 μV^2 (s.d. $\pm$ 3.6 μV^2 ; range 6–18 μV^2). Power of ROA in delta brushes showed a significant difference between the PT' and FT' groups ($p < 0.05$).

4. Discussion

We found that power of ROA in delta brushes on neonatal EEG was higher in PT than in FT. ROA power and amplitude is lower in more mature neonates. Physiological oscillatory activity of frequency up to 40 Hz can be observed from 8 months gestational age in the neonatal brain [6]. These fast oscillations were reported in relation to formation of somatotopic organization and neural networks [4]. Delta brushes were more frequently seen in PT than FT in our study. MBFA showed that power of ROA in delta brushes was higher in PT than FT and that frequency range of ROA was similar. The amplitude of delta brushes in younger neonates may correlate with developing neural networks.

Phenobarbital exposure did not have a significant impact on ROA in delta brushes among maturing neonatal EEGs. Phenobarbital acts as an allosteric inhibitor on the GABA_A receptor linked chloride channel by prolonging chloride channel opening. GABA_A receptors are theorized to play a role in generating oscillatory activity in both rat model and human brain. In the neonate under-expression of the KCC2 chloride channel results in depolarization rather than the expected hyper-polarization [7].

Confounding factors should be considered in evaluating the observed differences in ROA power. In our study we observed significantly fewer delta brushes with increasing post-conceptual age. Delta brushes are known to be less apparent with increasing age and not seen in the normal EEG beyond the age of 44 weeks [1]. The presence of delta brushes after this age is taken as an indicator of cerebral dysmaturity. The observed reduction in ROA power with increasing age is a real reflection of changes in cerebral activity. Apart from the explanation of development of neural networks in the neonatal brain, there may be increasing skull thickness with increasing post-conceptual age. This may be particularly true in the pre-term subjects in the PT and PB group, whose EEGs were performed at between 31 and 36 weeks post-conceptual age.

Ictal characteristics of neonatal seizures have been well-studied. The characteristics of neonatal seizures changes with maturation and term neonates have longer seizures than pre-term neonates (14.2 min vs 3.1 min), with a larger proportion having status epilepticus (33% vs 9%) [8]. We reported that ictal duration and sustained power on EEG of neonates who develop post-neonatal seizures differed from those without post-neonatal seizures [9]. Severe background abnormalities such as marked attenuation and asymmetry and excessive/focal sharp activities are associated with a high risk of seizures in the subsequent 18–24 h [10]. However, neonatal EEG recordings lack a clear equivalent to interictal epileptiform discharges in the adult and pediatric EEG recordings [2]. Comparison of ROA frequency and power on MBFA in epileptic hemimegalencephalic side and normal hemisphere has demonstrated distinct faster activities in the interictal discharges in hemimegalencephaly neonates and infants [15]. ROA may have potential as an interictal marker of seizure susceptibility. High frequency oscillations can be physiological or pathological [11–14]. Attempts to delineate the differences between pathological and physiological activity have been based on frequency, location and mechanism [5]. Additional characterisation of ROA has the potential to aid discernment between physiological and pathological oscillatory activities. MBFA provides a modality to aid characterisation of ROA in neonatal interictal EEG.

5. Conclusion

MBFA helps characterise ROA in the neonatal EEG. This technique may aid evaluation of ROA of delta brush in maturing neonates and may provide additional information to supplement frequency and anatomical location in the assessment of risks for future seizure recurrence and developing epilepsy.

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